

Yuhan Cao

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🌐 <https://teafrogsf.github.io/>

Research Experience

- 2025 –  **Research Intern**, Berkeley RDI Center, University of California, Berkeley.
Advisor: Prof. Dawn Song
Projects: Agents; LLM Alignment; RL for Generalization
- 2024 – 2025  **Research Intern**, Shanghai Qi Zhi Institute & Tsinghua University.
Advisor: Prof. Tianxing He
Projects: Mechanism Design for LLM Providers;
Benchmarking LLM for Test Case Generation in Competiton Programming
- 2024 – 2024  **Research Intern**, Microsoft Research Asia.
Advisor: Dr. Dongsheng Li and Dr. Yansen Wang
Project: AI for Drug Seeking through EEG (Brain Science)
- 2023 – 2023  **Research Intern**, daGAME Lab, Peking University.
Advisor: Prof. Xiaotie Deng and Dr. Zhaohua Chen
Project: Liquid Welfare on Truthful Ad Auctions
- 2022 – 2025  **Undergraduate Researcher**,
The ShanghaiTech Multi-Agent systems Research Team (SMART),
ShanghaiTech University.
Advisor: Prof. Dengji Zhao
Projects: Combinatorial Auctions on Social Networks; Double Auctions on Social Networks

Education

- 2020 – 2025  **B.Eng. in Computer Science**, ShanghaiTech University
Thesis: *Mechanism Design in Large Language Model Scenario*.
- 2023 – 2023  **Visiting Student in Computer Science Thematic Track**,
University of Wisconsin-Madison.

Highlighted Research

The symbol * means equal contribution, and † means corresponding author.

-  Agents' Last Exam, Y. Sun*, X. Han*, W. Zhang*, Y. Pang*, T. Wang*, **Y. Cao***, Y. Huang*, C. Duroiu, H. Zhang, J. Lin, et al., 2026. arXiv: 2606.05405 [cs.AI]. 🔗 URL: <https://arxiv.org/abs/2606.05405>
We introduce Agents' Last Exam (ALE), a living benchmark for evaluating AI agents on long-horizon, economically valuable, real-world tasks with verifiable outcomes. Developed with 250+ industry experts, ALE covers 1K+ tasks across 55 sub-fields and 13 industry clusters aligned with O*NET/SOC 2018. Current results show the hardest tier remains far from saturated, with average full pass rates below 1%.
-  RL Grokking Recipe: How Does RL Unlock and Transfer New Algorithms in LLMs? Y. Sun, **Y. Cao**, P. Huang, H. Bai, H. Hajishirzi, N. Dziri†, and D. Song†, *International Conference on Learning Representations*, 2026. arXiv: 2509.21016 [cs.LG]. 🔗 URL: <https://arxiv.org/abs/2509.21016>
We introduce DELTA-Code, a controlled benchmark of synthetic coding problem families designed to probe whether LLMs can acquire genuinely new reasoning strategies through RL. We discover a striking grokking phase transition: after an extended period with near-zero reward, RL-trained models abruptly climb to near-perfect accuracy. We explore key training ingredients—staged warm-up, experience replay, curriculum training, and verification-in-the-loop—and evaluate transferability along exploratory, compositional, and transformative axes.

- 3 Demonstrations of Integrity Attacks in Multi-Agent Systems, C. Zheng*, **Y. Cao***, X. Dong, and T. He†, 2025. arXiv: 2506.04572 [cs.CL].  URL: <https://arxiv.org/abs/2506.04572>
We explore integrity attacks in LLM-based multi-agent systems, where a malicious agent uses subtle prompt manipulation to bias system operations for self-interest without disrupting core functionality. We examine four attack types—Scapegoater, Boaster, Self-Dealer, and Free-Rider—and show that strategically crafted prompts can bypass advanced LLM-based monitors such as GPT-4o-mini and o3-mini.

Other Publications and Manuscripts

- 1 Can Large Language Models Reinvent Foundational Algorithms? J. Zhao, H. Luo, Y. Wang, **Y. Cao**, P. Sheng, and T. He, 2026. arXiv: 2604.05716 [cs.AI].  URL: <https://arxiv.org/abs/2604.05716>
- 2 **[Oral]** Pay for The Second-Best Service: A Game-Theoretic Approach Against Dishonest LLM Providers, **Y. Cao***, Y. Wang*, S. Liu, M. Li, Y. Tao, and T. He†, *The Web Conference*, 2026. arXiv: 2511.00847 [cs.GT].  URL: <https://arxiv.org/abs/2511.00847>
- 3 AICrypto: A Comprehensive Benchmark for Evaluating Cryptography Capabilities of Large Language Models, Y. Wang*, Y. Liu*, L. Ji*, H. Luo*, W. Li, X. Zhou, C. Feng, P. Wang, **Y. Cao**, G. Zhang, et al., *International Conference on Machine Learning*, 2026. arXiv: 2507.09580 [cs.CR].  URL: <https://arxiv.org/abs/2507.09580>
- 4 “Gain as a Loser”: Incentivizing Losers in Diffusion Combinatorial Auctions, X. Li*, M. Li*, **Y. Cao***, and D. Zhao†, *Under Review*, 2026.
- 5 Can LLMs Generate Reliable Test Case Generators? A Study on Competition-Level Programming Problems, **Y. Cao***, Z. Chen*, K. Quan, Z. Zhang, Y. Wang, X. Dong, Y. Feng, G. He, J. Huang, J. Li, et al., *Under Review*, 2025.
- 6 Distributed Brain Network Activity Drives Suppression of Methamphetamine Seeking by DBS and TMS, **Y. Cao***, W. Zhang*, X. Zhang, Y. Wang, K. Ren, R. Zhai, H. Jiang, Z.-Q. Xiong, and M. Zhao, *Manuscript*, 2025.
- 7 Convince Me with More Paragraphs: Generate and Evaluate Persuasive Writing with Multi-Agent LLMs, Z. Ling*, Y. Zhou*, **Y. Cao*†**, J. Zhang, W. Lyu, X. Liu, and H. Zhang, *Under Review*, 2025.
- 8 **[Oral]** Double Auction on Diffusion Network, M. Li, **Y. Cao**, and D. Zhao†, in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, 2024, pp. 9848–9855.

Selected Awards

2022		Merit Student (3%~7%) , ShanghaiTech University
2023		Student Scholarship (~7200 USD) , ShanghaiTech University
2022,2024,2025		Outstanding Teaching Assistant , ShanghaiTech University
2024		2nd Prize , ICPC Hongkong Regional
2022		1st in Honorable Mention , ICPC EC Final

Service

2026		Reviewer , NeurIPS 2026
		Reviewer , ICLR 2026 AIMS
2025		PC Member/Reviewer , WWW 2026
		Reviewer , ARR May 2025
2024		Student Volunteer , AAAI 2024